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Surname:															
First Name(s):															

Department of Mathematics and Applied Mathematics
MAM2000W: 2LA (Linear Algebra)

June exam	28 May 2019
Time : 2 hours	Maximum marks: 68
	Full marks: 67

- This question paper consists of 13 pages (including this one).
- Answer all questions in the spaces provided on this question paper. Use the backs of pages when you run out of space.
- Use your *UCT Examination Answer Book* for rough work. We won't take this work in! Only the answers that you write on this question paper will be marked.
- Calculators are not allowed. When your answer contains constants such as π , e or $\sqrt{2}$, leave them in that form. Don't simplify your answers.
- Be careful to provide answers that we can read and make sense of at all times. We will pay attention to your presentation as well as the content. Work that is poorly presented will be penalized. If we can't read your handwriting, we will mark your solution incorrect.

DO NOT WRITE BELOW THIS LINE

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	TOTAL
17	4	6	10	11	10	5	5	68

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Question 1. [17 points] Let $A = \begin{pmatrix} 3 & 2 & 1 \\ 0 & 5 & 0 \\ 6 & 1 & 2 \end{pmatrix}$.

Write the answers to the following questions in the blocks provided. **Do not show working!!!**

a. $\det(A) =$

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 . (1)

b. Is it possible to write A as a product of elementary matrices? Give a reason for your answer that does not require any further calculation. (1)

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c. Find elementary matrices E_1 and E_2 and an upper triangular matrix U such that $A = E_2E_1U$. (3)

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d. Find a basis for the nullspace of A . (1)

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e. Find a basis for the column space of A . (1)

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- f. Find a basis for the row space of A . (1)

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- g. The rank of A is , and its nullity is . (1)

- h. Find all the eigenvalues of A , together with their multiplicities. (2)

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- i. Can you say, without further calculation, whether A is diagonalizable? Give a reason for your answer. (1)

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- j. For each eigenvalue of A , find the corresponding eigenspace. (4)

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- k. Is A diagonalizable? Give a reason for your answer. (1)

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Question 2. [4 points] Let A be an $m \times n$ matrix. Complete each of the following statements by choosing one of the options (A) - (G) below. You may use an option more than once, and you do not have to use all options.

a. If $A\mathbf{x} = \mathbf{b}$ is consistent for all $\mathbf{b} \in \mathbb{R}^m$, then

b. If A is non-singular, then

c. If A is singular, then

d. If $m = n$ and A is invertible, then

(A) the rows of A are linearly independent.

(B) the columns of A are linearly independent.

(C) the rows of A span $\mathbb{R}^n$.

(D) the columns of A span $\mathbb{R}^m$.

(E) $\text{Nullity}(A) > 0$.

(F) A is diagonalizable.

(G) 0 is not an eigenvalue of A .

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Question 3. [6 points] Complete the following:

- a. If A is a 2×2 matrix with determinant 6 and trace 5, then its eigenvalues are... (1)

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- b. Let $T : P_3 \rightarrow P_3$ be a linear transformation for which

$$T(x^3) = x^2, \quad T(x^2) = x^2, \quad T(x) = x, \quad T(1) = x.$$

Then $T(5x^2 + 3) =$

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and $T(0) =$

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(2)

- c. Let $B = \begin{pmatrix} 7 & 4 \\ -8 & -5 \end{pmatrix}$. You are given that

$$B \begin{pmatrix} 1 \\ -1 \end{pmatrix} = 3 \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad \text{and} \quad B \begin{pmatrix} 1 \\ -2 \end{pmatrix} = - \begin{pmatrix} 1 \\ -2 \end{pmatrix}.$$

Find an invertible matrix P which diagonalizes B , and the resulting diagonal matrix D . (2)

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Write down an expression relating the matrices B , P and D . (1)

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Question 4. [10 points]

In P_3 , let $S = \{x^3 + x, x^2 + 2x - 1, x, x^2 - 1\}$.

Prove or disprove each of the following statements. **Show all your working!**

a. S spans P_3 .

(4)

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b. S is linearly independent

(2)

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c. S is a basis for P_3 .

(1)

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d. $\dim(\text{span}(S)) = 3$.

(3)

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Question 5. [11 points]

For each of the following statements, write **True** or **False**, and justify your answer. (No marks will be given for just writing “True” or “False” with no explanation, or with an incorrect explanation.)

- a. Suppose that after Gauss reducing the augmented coefficient matrix of a system of linear equations, we get the following matrix:

$$\left(\begin{array}{ccc|c} 1 & -2 & 0 & 4 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

The solution of this system is a plane in $\mathbb{R}^3$. (2)

- b. If B is a non-zero matrix and $B = E_k \cdots E_1 U$ where $E_1, \dots, E_k$ are elementary matrices and U is in row echelon form, then B is invertible. (2)

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- c. The set of all $A \in M_{2 \times 2}(\mathbb{R})$ such that $A^2 = \mathbf{0}$ is closed under scalar multiplication. (2)

- d. The set of all $A \in M_{2 \times 2}(\mathbb{R})$ such that $A^2 = \mathbf{0}$ is a subspace of the real vector space $M_{2 \times 2}(\mathbb{R})$. (2)

- e. If A and B are $n \times n$ matrices such that $AB = 0$, then $\text{rank}(A) + \text{rank}(B) \leq n$. (3)

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Question 6. [10 points] Read the theorem and its proof below and then answer the questions that follow.

Theorem If $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m$ are eigenvectors of a square matrix A associated with distinct eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_m$ respectively, then $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}$ is linearly independent.

Proof: Suppose that S is not linearly independent. We certainly know that $\{\mathbf{v}_1\}$ is linearly independent, so let $k > 1$ be the first integer such that $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is linearly dependent. Then there are scalars $a_1, a_2, \dots, a_k$, not all zero, such that

$$a_1 \mathbf{v}_1 + a_2 \mathbf{v}_2 + \dots + a_k \mathbf{v}_k = \mathbf{0}. \quad (1)$$

If we multiply both sides of (1) on the left by A , we get

$$a_1 \lambda_1 \mathbf{v}_1 + a_2 \lambda_2 \mathbf{v}_2 + \dots + a_k \lambda_k \mathbf{v}_k = \mathbf{0}. \quad (2)$$

Now multiply (1) by λ_k :

$$a_1 \lambda_k \mathbf{v}_1 + a_2 \lambda_k \mathbf{v}_2 + \dots + a_k \lambda_k \mathbf{v}_k = \mathbf{0}. \quad (3)$$

If we now subtract (2) from (3) we get:

$$a_1(\lambda_k - \lambda_1) \mathbf{v}_1 + a_2(\lambda_k - \lambda_2) \mathbf{v}_2 + \dots + a_{k-1}(\lambda_k - \lambda_{k-1}) \mathbf{v}_{k-1} = \mathbf{0}. \quad (4)$$

From this we can deduce that

$$a_1(\lambda_k - \lambda_1) = a_2(\lambda_k - \lambda_2) = \dots = a_{k-1}(\lambda_k - \lambda_{k-1}) = 0. \quad (5)$$

Therefore we must have $a_1 = a_2 = \dots = a_{k-1} = 0$. (6)

Going back to equation (1) this leaves us with $a_k \mathbf{v}_k = \mathbf{0}$ and from this we can deduce that $a_k = 0$. (7)

But this gives a contradiction, (8)

so the set S must be linearly independent.

a. How do we know $\{\mathbf{v}_1\}$ is linearly independent? (2)

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b. Explain carefully how equation (2) follows from equation (1). (2)

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- c. In deducing (5) from (4), which property of the set $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{k-1}\}$ are we using? How do we know it has this property? (2)

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- d. Why can we deduce at (7) that $a_k = 0$? (1)

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- e. What is the contradiction referred to at (8)? (1)

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- f. Where in the proof is the fact that the eigenvalues are distinct used, and how? (2)

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Question 7. [5 points] Prove the following theorem.

Theorem: Let $S = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ be a non-empty set of vectors in a vector space V . If S is linearly independent, and if $\mathbf{v}$ is a vector in V that is not in the span of S then $S \cup \{\mathbf{v}\}$ is still linearly independent.

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Question 8. [5 points] For a square matrix A , let $\sigma(A)$ denote the set of all eigenvalues of A . Prove that if A and B are square matrices of the same size, then $\sigma(AB) = \sigma(BA)$.